Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Luz Sotelo	Email	coria-consortium@purdue.edu
Institute Name	Purdue University			
Address	2550 Northwestern Ave, west Lafayette, IN 47906			

2. Technology Overview

Title of Proposed Technology	**Development of AI-Based Hybrid Fault Prediction System Using Acoustic and Electronic Operational Data for Agricultural Machinery**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials			
	Medium Category	Small Category	Medium Category	Small Category
	Industrial / General Machinery	*Agricultural Machinery*		
Concept of Proposed Technology	**Core Concept** The proposed technology aims to establish an AI-based hybrid fault prediction system for agricultural machinery by integrating electronic diagnosis with sound-based machine intelligence to detect early-stage mechanical anomalies. While current Controller Area Network (CAN)-based diagnostic platforms provide valuable insights into the electronic nervous system of machinery, they remain blind to the physical degradation mechanisms—bearing wear, gear chipping, hydraulic cavitation—that often precede catastrophic failure. The proposed technology introduces machine listening using acoustic spectrogram analysis and unsupervised AI to detect early-stage mechanical anomalies that conventional electronic sensors cannot capture. **Key Functions and Potential Applications**			

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	Key Functions ● Acoustic Anomaly Detection ■ The system detects subtle mechanical faults—bearing wear, gear tooth defects, cavitation in hydraulic pumps—that are virtually undetectable through CAN sensors alone. ■ By capturing and analyzing the unique sound patterns associated with emerging mechanical failures, the solution eliminates blind spots inherent in existing diagnostic systems. ● Non-contact / Retrofit Hardware Module ■ Using diaphragm-based acoustic pickup and localized acoustic hose mechanisms, the system captures high-fidelity operational sound without modifying machine structures. ■ A compact microphone module mounted on the telematics unit collects target frequencies using a stethoscope-like interface, enabling easy deployment across both new and aging fleets. ■ This retrofit-friendly approach dramatically reduces installation barriers and accelerates field adoption. ● Edge–Cloud Hybrid Architecture ■ A lightweight edge device focuses solely on noise acquisition and uploading, while AI-based spectral analysis and anomaly detection are performed in the cloud using scalable computing resources. ■ This architecture reduces hardware costs, allows rapid algorithm updates, and supports real-time fault prediction across diverse machinery platforms. **Potential Applications** ● Agricultural tractors, combines, sprayers, and utility vehicles requiring predictive maintenance ● Legacy machinery markets where electronic diagnostic coverage is limited ● OEM and aftermarket retrofit solutions enabling sound-based health monitoring ● Remote fleet management services leveraging cloud-based acoustic intelligence ● Subscription-based maintenance services (SaaS) centered on fault prevention
Rationale for Joint R&D Project	**Critical Technical Barriers** ● Limited Visibility of Mechanical Degradation Using Electronic Data Alone Existing agricultural machinery diagnostics rely heavily on CAN-based electronic signals, which accurately capture electrical and control-related faults but fail to detect early mechanical degradation. Subtle physical abnormalities—such as bearing fatigue, micro-chipping of gears, or hydraulic pump cavitation—produce no identifiable electronic signatures until after damage has progressed. Overcoming this blind spot requires developing a complementary sensing modality capable of identifying precursor-level mechanical anomalies. ● Absence of Fault Data for AI Model Training Mechanical failure events in agricultural machinery are rare, unpredictable, and highly

variable across machine types, operating environments, and wear conditions. This makes it nearly impossible to collect sufficiently labeled fault datasets for supervised learning. A major barrier is designing an AI model capable of high diagnostic accuracy despite the scarcity of ground-truth fault samples, necessitating unsupervised or one-class learning approaches that can operate using only normal-condition data.

- Challenges in Acoustic Data Acquisition Under Harsh Operating Environments

  The engine compartments of tractors and other agricultural machinery exhibit extreme noise, vibration, dust, and temperature variations. Conventional microphones cannot isolate fault-relevant frequencies from broadband ambient noise. A key barrier lies in engineering a rugged, selective acoustic sensing module—using diaphragms, localized acoustic channels, or structural filtering—that can reliably capture mechanical sound signatures without structural modification of the machine.

- Fusion of Heterogeneous Data Streams (Acoustic + Electronic) in Real Time

  Integrating acoustic features with CAN-based operational parameters requires precise time synchronization and robust fusion algorithms. Mechanical anomalies may only appear under specific load or RPM conditions, and failure to align these contexts can result in false alarms or loss of diagnostic sensitivity. Developing scalable feature-level fusion logic and context-aware detection models is a significant technical challenge.

- Edge–Cloud Scalability and Real-time Processing Constraints

  The system must process large volumes of acoustic data while minimizing hardware cost and power consumption at the edge device. Transmitting raw or semi-processed sound data to the cloud requires optimized compression, bandwidth management, and latency control. At the same time, the cloud infrastructure must support real-time spectrogram analysis and anomaly detection across thousands of distributed machines. Ensuring seamless end-to-end performance poses engineering, computational, and algorithmic barriers.

Why Global Collaboration?

- Cutting-edge Agricultural Research Infrastructure

  A global partnership is essential because collaboration involves an institution recognized as one of the world's leading agricultural research universities. Supported by the vast farmlands of the U.S. Midwest, Purdue University conducts top-tier agricultural engineering research and operates the Agronomy Center for Research and Education (ACRE)—a comprehensive research farm equipped for advanced studies in machinery dynamics, environmental conditions, and operational variability. Joint research with such a globally recognized research environment provides scientific credibility, following up cutting-edge research, and a foundation for developing technologies that is relable across international markets.

- Benchmarking of foundational IP and leading to foreground IP

  Purdue possesses multiple foundational technologies, which are officially recognized for their technical merit by the U.S. Patent and Trademark Office (USPTO), related to the sound-

	based mechanical diagnostics. Joint R&D with such a leading research institution enables the Korean SMEs to strengthen its technological leadership abroad. The experience, methodologies, and tacit knowledge accumulated during the development of the patents are critical assets that cannot be replicated through documentation alone. Global collaboration enables Korean partners to internalize the technical rationale behind the patented acoustic diagnostic framework and practical know-how in building scalable and noise-resilient acoustic AI models.
Key Words	1. Predictive Maintenance 2. Acoustic Intelligence 3. Electrical Signal Analytics 4. Multi-modal Sensor Fusion 5. Non-invasive Retrofit Module

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	**R&D Objectives** The project aims to establish an integrated hybrid diagnostic framework that combines electronic CAN-based monitoring with sound-based machine intelligence to enable early detection of mechanical anomalies in agricultural machinery. The overarching objective is to build a unified system encompassing acoustic sensing hardware, spectrogram-based AI analytics, and multi-modal sensor fusion algorithms that together provide reliable, real-time fault prediction across diverse machinery platforms. The final goal is to develop a deployable hybrid diagnostic solution that can capture fault-relevant mechanical signatures, analyze them through cloud-based AI models, and deliver accurate, context-aware predictions that complement and extend the capabilities of existing electronic diagnostic systems. **Detailed Objectives** 1. Development of a Robust Acoustic Sensing Hardware Module The first objective is to engineer an industrial-grade acoustic sensing module that can operate reliably in the extreme environment of agricultural machinery engine compartments—characterized by intense vibration, high temperatures, and broad-spectrum noise. This includes developing: ■ A rugged microphone assembly capable of isolating fault-relevant frequency bands ■ A selective housing structure that suppresses ambient noise while amplifying target mechanical signatures High-quality acoustic data is fundamentally linked to AI diagnostic performance, making this hardware the foundational component that ensures the accuracy and stability of the entire hybrid fault prediction system.			

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2. Cloud-Operable Spectrogram AI Model Optimized for Commercial Deployment

The second objective is to adapt and optimize the spectrogram-based anomaly detection model so that it can run efficiently on commercial cloud infrastructure using data uploaded from low-cost telematics devices. This involves:

- Applying quantization, pruning, and model compression techniques to reduce computational load
- Ensuring real-time inference capability on cloud servers without requiring high-end edge hardware
- Achieving seamless integration with existing telematics units to maintain retrofit market competitiveness

By enabling high-performance AI diagnostics without hardware upgrades, this objective directly supports rapid commercialization and cost-effective deployment across large fleets.

3. Implementation of a Hybrid Diagnostic Algorithm Combining Acoustic and CAN Data

Acoustic-only diagnostics risk false alarms due to unpredictable environmental noises typical in real agricultural operations. The third objective is to develop a hybrid sensor-fusion algorithm that synchronizes and integrates acoustic features with CAN parameters—such as engine RPM, load ratio, and coolant temperature—to achieve context-aware fault detection. Key tasks include:

- Real-time temporal alignment of heterogeneous data streams
- Feature-level fusion for enhanced robustness
- Learning condition-specific anomaly patterns to distinguish true mechanical failures from incidental noise

This hybrid approach significantly reduces false positives, maximizes diagnostic reliability, and ensures user trust in the final system.

R&D Target Specifications

Key Performance Indicators	Target	Weight	Measurement and Evaluation Method
Abnormal Noise Detection Accuracy	≥ 96%	25%	Evaluate model performance using normal and faulty acoustic datasets
CAN–Acoustic Fusion Diagnostic Accuracy	≥ 98%	25%	Verify accuracy improvement compared to acoustic-only models after integrating CAN data
Fault Prediction Lead Time	≥ 5 hours before failure	25%	Measure average time the system issues a warning prior to actual mechanical failure
AI Inference Speed	≥ 100 ms	25%	Measure latency from acoustic input to diagnostic output

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R&D Plan	**R&D pipeline** The R&D plan follows a structured pipeline that builds the full hybrid diagnostic system from hardware to AI modeling. The development is organized into six core pillars: 1. Development of Hardware for CAN and Acoustic Data Acquisition ■ Develop rugged sensing hardware capable of capturing high-fidelity acoustic signals and CAN-based electronic data under real agricultural operating conditions. ■ Integrate microphones, housings, connectors, and power modules into a field-deployable data acquisition unit. 2. Development of Embedded Data Acquisition Software ■ Implement embedded software pipelines for real-time data routing, buffering, timestamping, and communication. ■ Ensure stable acquisition of synchronized CAN and acoustic streams through optimized on-device data handling. 3. Construction of Time-Synchronized and Labeled Multi-Modal Datasets ■ Establish precise temporal alignment mechanisms for CAN and acoustic data across varying machine states such as RPM, load, and duty cycles. ■ Build labeled datasets that capture operational contexts needed for hybrid diagnostic model training and validation. 4. Development of Data Pre-processing Pipelines ■ Develop preprocessing workflows including filtering, segmentation, normalization, and windowing to ensure consistent data quality. ■ Extract preliminary time–frequency features and isolate component-specific acoustic signatures. 5. Development of Acoustic Analysis and Spectrogram Processing Principles ■ Analyze with frequency-band to capture mechanical fault-related patterns and generate spectrograms. ■ Formulate feature extraction principles tailored to bearings, gears, and hydraulic components. 6. Development of Unsupervised AI Models for Anomaly Detection and Clustering ■ Build autoencoder-based and clustering-based models that learn normal behavior and detect deviations without fault labels. ■ Evaluate anomaly scores, reconstruction errors, and cluster separation metrics to derive reliable early-warning indicators.

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4. R&R (Roles and Responsibilities) 역할 및 책임

Expected R&R of Research Institute	The research institute is responsible for development of acoustic/CAN interpretation principles, diagnostic algorithms, spectrogram analysis workflows, and AI-based anomaly detection models. 1. Acoustic & CAN Data Interpretation and Diagnostic Algorithm Development ■ Develop analytical principles for acoustic signal interpretation, frequency-band analysis, and fault-related acoustic pattern extraction. ■ Establish hybrid diagnostic logic frameworks combining mechanical signatures with electronic operational parameters. 2. Spectrogram Processing and Feature Engineering ■ Formulate STFT-based spectrogram generation methods tailored to agricultural machinery. ■ Develop component-specific feature extraction techniques for bearings, gears, hydraulic pumps, and rotating components. 3. AI Modeling and Unsupervised Learning Development ■ Build unsupervised AI models (autoencoders, clustering frameworks) that learn normal operational behavior and detect anomalies without fault labels. ■ Optimize model performance using multi-modal datasets and evaluate reconstruction error metrics, anomaly scores, and cluster separability. 4. Theoretical Validation and Algorithmic Refinement ■ Conduct iterative algorithm validation using lab-level benchmark datasets. ■ Provide methodological guidance for integrating acoustic features into sensor-fusion workflows.
Expected R&R of Korean SME Partner	The Korean SME leads the development of hardware, embedded software, real-machine data acquisition, synchronization algorithms, system integration, and user-interface deployment. 1. Hardware Design and Field-Deployable Data Acquisition Systems ■ Design and fabricate rugged acoustic–CAN data acquisition hardware suitable for agricultural machinery environments. ■ Integrate microphones, housings, connectors, power systems, and mounting structures into deployable units. 2. Embedded Software and Synchronization Algorithms ■ Develop embedded pipelines for real-time acoustic and CAN data routing, buffering, timestamping, and communication. ■ Implement synchronization logic ensuring precise temporal alignment of CAN and acoustic signals under variable machine states. 3. Construction of Multi-Modal Datasets Through Real-Machine Testing ■ Conduct field data collection using actual agricultural machinery under diverse operating conditions. ■ Build labeled datasets capturing operational contexts required for training and validating hybrid diagnostics.

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	4. System Integration and Cloud/Telematics Architecture Development ■ Integrate preprocessing pipelines, spectrogram generation modules, and AI inference engines into the telematics–cloud ecosystem. ■ Ensure seamless end-to-end operation from sensor input to cloud-based analysis. 5. User Interface and Service Platform Development ■ Develop intuitive web/mobile dashboards for delivering diagnostic information, anomaly warnings, and maintenance insights to end users. ■ Implement real-time monitoring functions with emphasis on usability, reliability, and field-ready serviceability.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	● Technological Outcomes ■ Diagnostics AI System for Agricultural Machinery Develop an AI-driven system that detects mechanical anomalies by combining acoustic signals with CAN-based electronic data, enabling early identification of fault-related patterns. ■ Establishment of Multi-Modal Predictive Maintenance Technology Build a diagnostic framework that integrates acoustic features with operational parameters from CAN data to improve fault prediction accuracy and reduce false alarms. ● Commercialization Potential ■ Retrofit Market Applicability Enable older and existing agricultural machinery to adopt AI-based diagnostic functions through a compact acoustic sensing module, supporting global aftermarket deployment. ■ Subscription-Based Service Model Provide continuous, farm-operation-focused predictive maintenance services that can be commercialized as a scalable SaaS offering. ● Economic and Industrial Impact ■ Reduced Downtime and Improved Productivity Lower unexpected equipment failures during peak operational periods, minimizing financial losses and improving overall agricultural output. ■ Expanded Export Opportunities and Stronger Industrial Collaboration Strengthen competitiveness of domestic agricultural solutions and support export growth through integration with machinery entering global markets.

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6. References

(1) Jun, M. B.-G., & Kim, H. (2023). *System and methods for machine anomaly detection based on sound spectrogram images and neural networks* (U.S. Patent No. 11,688,415 B2). U.S. Patent and Trademark Office. https://patents.google.com/patent/US11688415B2